

Economic Impact of the RCEP

Ex post evidence and ex ante CGE simulations

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Why evaluate RCEP now?

- RCEP entered into force on January 1, 2022 (RCEP, 2020)
 - ASEAN, Australia, China, Japan, Korea, and New Zealand
 - Around 30% of world population and GDP before enactment (United Nations, 2024; IMF, 2025)
- The agreement covers not only tariffs, but also services, investment, rules of origin, e-commerce, and regulatory cooperation
- Evaluation is timely because the mandated review approaches in 2027
 - What has changed in the first years?
 - Where could deeper implementation generate gains?

Related literature

- Early RCEP studies evaluate potential gains using CGE models under prospective liberalization scenarios
 - e.g., Cheong and Tongzon (2013), Lee and Itakura (2018), and Petri and Plummer (2020)
- More recent studies incorporate negotiated commitments and broader channels such as services, logistics, investment, and financial frictions
 - e.g., Itakura (2022) and Jin et al. (2025)
- Evidence using post-entry data remains limited because RCEP has only recently entered into force
 - This study links early ex post evidence with ex ante CGE simulations

This study

- We provide a quantitative evaluation of RCEP using two complementary approaches
 - **Ex post**: structural gravity estimation using observed trade, inward FDI, RTA, BIT, and tariff data
 - **Ex ante**: dynamic CGE simulations of deeper RCEP implementation
- We connect the two approaches
 - Gravity estimates provide targets for non-tariff barrier and investment shocks
 - CGE simulations quantify medium-run macroeconomic and trade effects
- Main message: RCEP's larger gains are likely to come from **services, NTBs, and investment facilitation**, not tariffs alone

Ex post analysis: objectives

- Estimate the average short-run effect of RCEP on:
 - goods imports
 - services imports
 - inward FDI positions
- Examine whether RCEP changed the relationship between tariffs and goods imports
 - Did imports among RCEP pairs become more responsive to tariff reductions?
- Follow standard structural gravity practice (Yotov et al., 2016)

Structural gravity specification

- Estimate the following gravity equation by PPML (Santos Silva and Tenreyro, 2006)

$$X_{ij,t} = \exp \left(\beta \text{RCEP}_{ij,t} + Z'_{ij,t} \delta + \gamma_{ij} + \pi_{i,t} + \psi_{j,t} \right) \cdot \varepsilon_{ij,t}$$

- $X_{ij,t}$: goods imports, services imports, or inward FDI positions
- $Z_{ij,t}$: controls for other deep/shallow RTAs and BITs (only for FDI)
- Country-pair FE absorb time-invariant bilateral factors, helping address endogeneity concerns in RTA estimates (Baier and Bergstrand, 2007)
- Exporter-year / importer-year FE absorb output, expenditure, country-year factors, and multilateral resistance terms (Anderson and van Wincoop, 2003)

Data for ex post estimation

- Goods trade: UN COMTRADE, importer-reported CIF values, 1996–2024
- Services trade: UN COMTRADE, bilateral services imports, 2000–2024
- Inward FDI: IMF CDIS positions, 2009–2023
- Tariffs: WTO Stats, MFN and preferential rates at HS6 level
 - applied tariff = minimum of MFN and preferential rates
 - aggregated to importer–exporter–year using import weights
- RTA depth: World Bank Deep Trade Agreements Database (Hofmann et al., 2017)
 - other RTAs with depth at least comparable to RCEP are classified as deep
- BITs: UNCTAD International Investment Agreements Navigator

Summary statistics by treatment status

Variables	RCEP = 1		RCEP = 0	
	Mean	Zero (%)	Mean	Zero (%)
log Goods Imports	21.16	0.18	14.20	30.01
log Services Imports	21.28	39.62	17.25	69.21
log Inward FDI	21.02	4.94	17.09	50.53
Other Deep RTA	0.86	–	0.09	–
Shallow RTA	0.12	–	0.10	–
BIT	0.41	–	0.21	–
Applied Tariff (%)	1.21	27.55	7.27	14.15

RCEP pairs are larger, more connected, and already face lower tariffs.

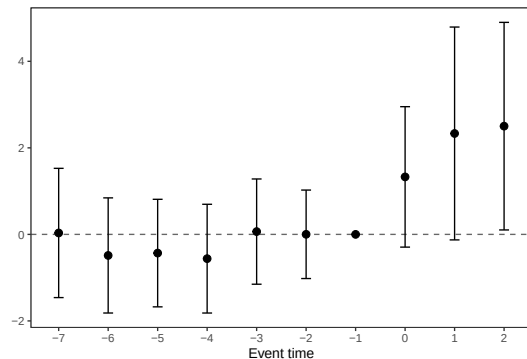
Main gravity results

Dependent variable	Goods imports	Services imports	Inward FDI
RCEP	-0.0891 (0.0583)	0.1814** (0.0880)	-0.0732 (0.0801)
Other deep RTA	0.0877*** (0.0263)	0.0622 (0.0446)	0.1455** (0.0658)
Shallow RTA	0.0505 (0.0707)	0.0143 (0.1318)	0.0370 (0.0873)
BIT			0.0068 (0.0090)
Observations	895,696	90,716	108,375
Pseudo R ²	0.991	0.982	0.989

Country-pair, importer-year, and exporter-year fixed effects included.

Additional ex post evidence: tariff responsiveness

- Tariff coefficients increase for intra-RCEP trade after entry into force
- No clear pre-trend is observed before RCEP entered into force
- Although the average goods-trade effect is not yet visible, RCEP may have induced a structural change in tariff responsiveness



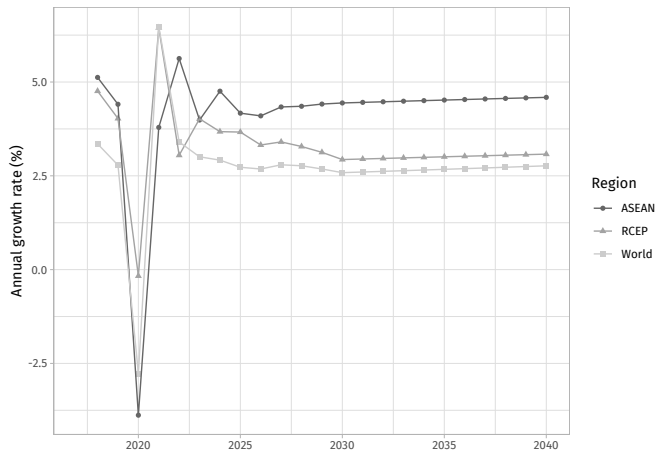
Why ex ante simulation?

- Ex post data capture only the first years after entry into force
- RCEP commitments are implemented gradually
- Potential gains may operate through:
 - reduced non-tariff barriers in goods trade
 - lower barriers in services trade
 - improved investment conditions and capital accumulation
- A recursive dynamic CGE model quantifies how these channels propagate through the economy

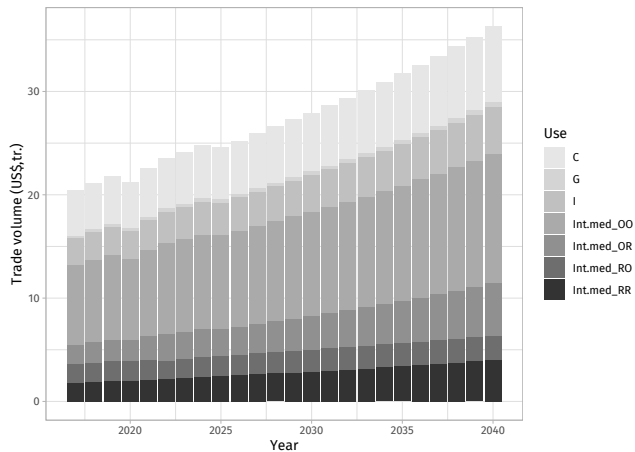
CGE model and database

- Model: dynamic GTAP model with global supply chain structure (Ianchovichina and Walmsley, 2012)
 - multi-region, multi-sector recursive dynamic CGE framework
 - Armington differentiation by origin
 - capital accumulation and international capital mobility
- Database:
 - GTAP v11, benchmark year 2017 (Aguilar et al., 2023)
 - OECD ICIO tables to distinguish intermediate and final uses (OECD, 2025)
 - aggregation: 25 countries/regions and 24 sectors
- Baseline path calibrated to population, GDP, and investment projections through 2040 (United Nations, 2024; IMF, 2025)

Baseline: projected real GDP growth



Baseline: world trade by use



RCEP policy scenarios

Baseline: business-as-usual path through 2040 without RCEP-specific components

S1: tariff reductions based on RCEP Annex I (RCEP, 2020)

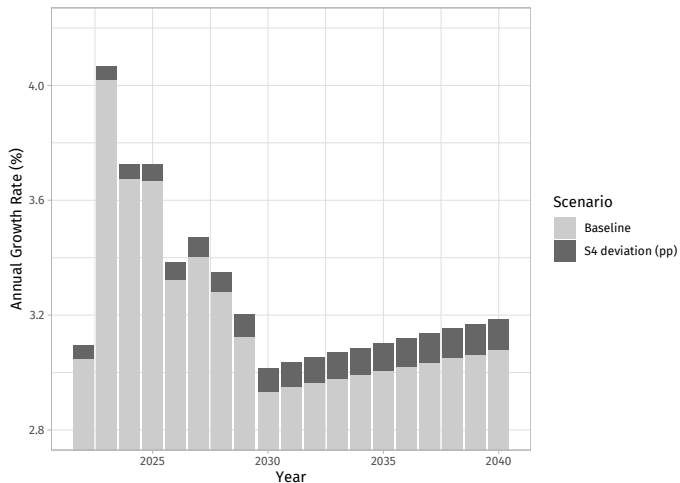
S2: S1 + lowering NTBs to goods imports

S3: S2 + reducing barriers to services trade

S4: S3 + facilitating inward FDI

- Goods NTB target: 9.2% import increase from deep RTA estimate
- Services target: 19.9% services import increase from RCEP estimate
- FDI target: 15.7% inward FDI increase from deep RTA estimate
- Deep-RTA estimates are used as upper-bound targets

Impact on RCEP real GDP growth



Economic impact of RCEP in 2040

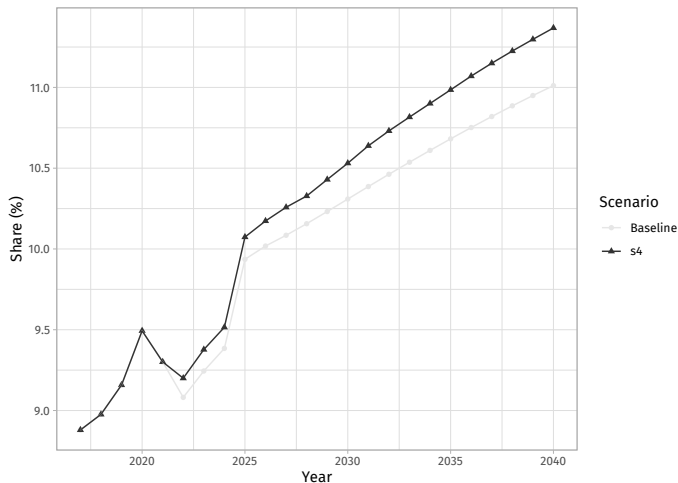
Indicator	Region	S1	S2	S3	S4
Real GDP (%)	RCEP	0.1	0.4	1.1	1.5
	ASEAN	0.1	0.4	1.1	1.8
	World	0.0	0.0	0.2	0.1
Investment (%)	RCEP	0.2	0.7	1.7	3.4
	ASEAN	0.2	0.5	1.4	3.4
	World	0.0	0.1	0.3	0.2
Welfare (%)	RCEP	0.1	0.4	1.0	1.3
	ASEAN	0.0	0.3	0.8	1.0
	World	0.0	0.1	0.2	0.2

Intra-RCEP trade and intermediate inputs in 2040

	Total		Intermediates	
	(%)	US\$ bn	(%)	US\$ bn
S1	3.5	203.5	3.3	132.6
S2	3.8	216.4	3.5	138.0
S3	4.2	240.0	3.7	149.8
S4	4.5	258.5	3.8	152.7

- Deeper RCEP implementation expands intra-regional trade
- Intermediate-input trade rises almost as strongly as total trade

Intra-RCEP intermediate trade share



Conclusion

- Ex post evidence:
 - limited short-run average effects on goods trade and inward FDI
 - clear early gains in services trade
 - stronger tariff responsiveness among RCEP pairs after entry into force
- Ex ante simulations:
 - tariff reductions alone generate modest gains
 - services NTBs and investment facilitation are major sources of medium-run gains
 - in S4, RCEP real GDP rises by 1.5% and welfare by 1.3% in 2040
- Policy implication:
 - realizing RCEP's potential depends on deeper implementation beyond tariffs

Appendix: country-level real GDP impact in 2040

Real GDP (%)					Real GDP (%)				
Country	S1	S2	S3	S4	Country	S1	S2	S3	S4
Indonesia	0.0	0.1	0.9	1.6	Japan	0.5	0.7	1.3	1.5
Malaysia	0.2	0.5	1.1	1.6	China	0.1	0.4	1.1	1.4
Philippines	0.1	0.4	1.0	1.7	Korea	0.5	0.7	1.3	1.3
Singapore	-0.1	0.0	0.9	1.9	Australia	0.0	0.1	0.9	2.2
Thailand	0.6	1.0	1.5	2.2	New Zealand	0.0	0.1	0.9	2.0
Viet Nam	0.5	1.2	1.6	3.0	RCEP	0.1	0.4	1.1	1.5